

Dhruv Venkat

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EDUCATION

University of Waterloo

Bachelor of Applied Science, Computer Engineering (Honours)

Sep 2024 - 2029

Waterloo, ON

TECHNICAL PROFICIENCIES

- **Languages:** TypeScript, Python, C++, Java, Go, SQL, RISC-V Assembly
- **Libraries and Frameworks:** Node.js, ReactJS, Next.js, Scikit-Learn, TensorFlow, Pandas, FastAPI
- **Cloud/DevOps:** AWS, Microsoft Azure, Git, Jenkins, PostgreSQL, GraphQL

PROFESSIONAL EXPERIENCE

Red Lane Group | *Software Development Intern*

May 2026 - Present

- Building an **AI orchestration pipeline** to classify, route, summarize, and escalate support tickets across 5+ products, using **Claude-powered agent workflows and custom tools**; projecting **100+ work hours saved/month**
- Modernizing legacy call transcription pipeline with a **real-time compliance recording bot to transcribe and summarize 100% of support agent calls**, reducing post-call processing latency by 20+ minutes by migrating to real-time streaming SDK transcription

Scotiabank | *Software Engineering Intern*

Sep 2025 - Dec 2025

- Designed and deployed AI-powered RAG agents using LangChain and FAISS to extract business requirements from user stories, **reducing requirements-gathering time by 40% and increasing development productivity by 25%**
- Implemented a **Tesseract OCR** and AI-driven parsing pipeline, **achieving a 60% reduction in manual compliance review time and 3× document throughput** by automating data extraction and signature verification of regulatory forms
- **Achieved 30% faster runtime and 50% reduction in maintenance effort for 8 enterprise applications** by upgrading backend platforms from Java 11/17 to Java 21, enabling long-term LTS support

Tangerine | *DevOps Engineering Intern*

Jan 2025 - Apr 2025

- Implemented a company-wide GitHub Copilot license tracker, **reducing monthly license spend by \$1,000+** by developing, deploying, and documenting an automated solution **using RESTful APIs in Apache Groovy**
- Deployed Python-based scanners **across 100+ GitHub repositories**, achieving a **20% reduction in high-severity vulnerabilities by deploying concurrent RHEL scripts** and automating detection and prioritization
- Designed and maintained **five multi-branch Jenkins CI/CD pipelines**, increasing build reliability by 30% through automated test suites, multi-branch strategies, and deployment orchestration

PROJECTS

flowstate - AI-Integrated IDE for the Terminal | [Link](#)

Apr 2026 - May 2026

C++, POSIX TTY, CMake, Git, OpenAI/Codex

- Architected a terminal-native code editor with **raw-mode input handling, custom rendering, redo/undo history, Git-aware change visualization, a built-in file picker, and AI-assisted features**
- Implemented **multi-language syntax highlighting, error detection, and project root resolution** via Language Server Protocol integrations for C/C++, Python, JavaScript/TypeScript, Go, and Rust
- Built an **integrated AI coding assistant** using Codex, enabling **editor-aware code generation**, contextual explanations, and **in-place code edits** directly in the terminal workflow